Exercises_Lecture1

June 28, 2023

1 Exercises

1. Print the result of the following operations using python:

   - 5 + 4 + 3 + 2 + 1
   - 5 * 4 * 3 * 2 * 1
   - 6^4

2. Use the math package to compute the following expression

   - $e^{\pi} - 1$

3. Let `a=True + True`. what is the type of the variable `a`?

4. Let `a=[1,2]` and `b = [5,1]` be two lists. what is the result of `c=3*a + b`?

5. Let `a=(6,8,3,0)` be a tuple.

- write a program that prints the second and last element of this tuple
- transform the previous tuple to a list and store the result in a variable called `b`.
- Insert the value `"A"` in the second position of the list `b`
- Delete the value `8` from the list
- Delete the first value of the list
- print the resulting list

6. Use pandas to open the three files contained in the data folder of this lesson (`data_cymbals.csv`, `data_invoices.xlsx`, and `sales_us.txt`).

7. Create a numpy array, `a`, with the numbers from 0 to 6.

8. Create a numpy array of length 10 with the squares of the first natural numbers, that is 1,4,9,...

9. Let `a=[1,2]` and `b = [5,1]` be two lists. Transform them to numpy arrays. What is now the result of `c=3*a + b`?

10. 4. Create an array of length 10 with the squares of the first natural numbers, that is 1,4,9,...

Optional Exercises

1. If `a=5.3` is a numeric variable (for instance an integer); then `(a*9)/3 == a*(9/3)` is a Boolean expression with value True. However, if `a="A"`, a string, is the same expression `(a*9)/3 == a*(9/3)` still a correct Boolean statement?

2. Let `names=["one", "two", "three"]` and `numbers=[1,2,3]` two lists. Construct two dictionaries, one in which the names are the keys and the numbers the values, and another in which the numbers are the keys and the names are the values

3. Create a numpy array of 200 elements in which the fist 100 elements are ones; and the last 100 elements are zeros

4. Create another array, `b`, with all even numbers from 50 to 60 (included).

5. open the fime `cymbals.csv` with pandas. How many measurements in the column **day1** are less than 440?